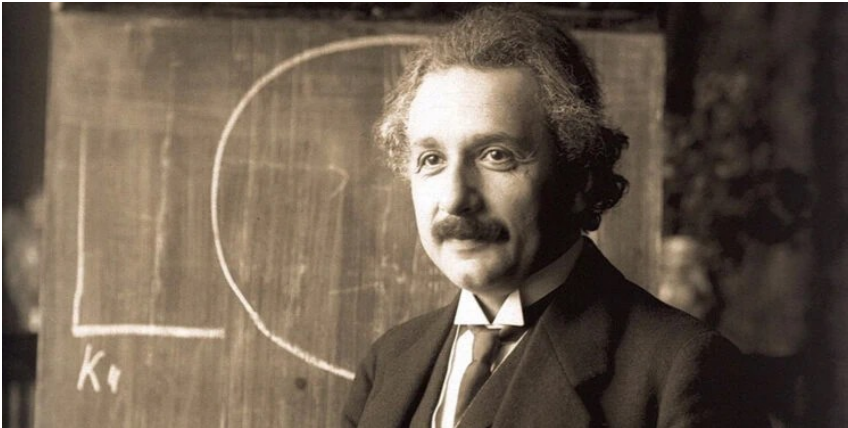


Albert Einstein's brain: What have scientists discovered?



For decades, scientists have sought to understand the biological foundations of genius by examining the physical characteristics of Albert Einstein's brain. Various studies indicate that while his brain weighed a normal amount, it

possessed distinct anatomical features and enhanced connectivity that may have contributed to his extraordinary cognitive abilities. These findings offer a glimpse into the potential neural substrates of his mathematical and visuospatial prowess.

Albert Einstein is widely recognized as one of the most influential physicists in history. In 1905, often referred to as his *annus mirabilis* or miraculous year, he published four groundbreaking papers that fundamentally altered the scientific understanding of the universe. These works addressed the photoelectric effect, Brownian motion, special relativity, and the equivalence of mass and energy. He later developed the general theory of relativity, cementing his reputation as a scientific revolutionary.

Because of his immense intellectual achievements, significant curiosity surrounded the source of [his genius](#). Researchers and the public alike questioned whether his abilities stemmed from his environment and education or if he possessed a unique biological advantage. This debate regarding nature versus nurture prompted a desire to analyze his physical brain for clues.

Einstein died on April 18, 1955, at Princeton Hospital due to a ruptured abdominal aortic aneurysm. During the autopsy, the pathologist Thomas Harvey removed Einstein's brain for scientific study. Harvey weighed the organ, photographed it from multiple angles, and preserved it using formalin. He subsequently sectioned the brain into approximately 240 blocks and prepared histological slides, which are microscopic slices of tissue used for studying cellular structure.

Anatomical Structure and the Parietal Lobes

In 1999, Sandra Witelson and her team at McMaster University published a study in [The Lancet](#) that focused on the gross anatomy, or visible structure, of the brain. They utilized photographs taken by Harvey in 1955 and measurements taken directly from the organ. They compared these against a control group of thirty-five male and fifty-six female brains.

Witelson found that Einstein's brain weight was not exceptional. However, the parietal lobes displayed striking differences. The study revealed that his parietal lobes were approximately 15 percent wider than those of the control group.

Furthermore, the researchers identified a unique pattern in the Sylvian fissure. This fissure is a deep fold that separates the temporal lobe from the frontal and parietal lobes. In typical brains, this fissure curves up and divides a region called the supramarginal gyrus.

In Einstein's brain, the Sylvian fissure followed an unusual trajectory that left the supramarginal gyrus undivided. Witelson proposed that this absence of division allowed for more efficient connections between neurons in that area. This region supports visuospatial cognition and mathematical thinking. The study suggests this unique anatomy may have facilitated the type of visual thinking Einstein used to develop his theories.

Surface Features and Motor Skills

In 2009, anthropologist Dean Falk published a study in [Frontiers in Evolutionary Neuroscience](#) that applied techniques from paleoanthropology to analyze photographs of Einstein's brain. This field usually focuses on the evolution of human anatomy, but Falk used these skills to identify previously overlooked details on the brain's surface.

Falk identified an unusual "knob-like" structure on the right motor cortex. The motor cortex is the area of the brain that controls voluntary muscle movement. This specific knob corresponds to the area that controls the left hand.

Falk noted that this feature is often seen in long-time musicians. Einstein famously played the violin from childhood. The study implies that this anatomical feature developed as a result of the intense, repetitive motion required to finger the strings with his left hand. This finding provides

evidence for plasticity, the brain's ability to physically change in response to repeated activity.

Falk also observed unusual grooving patterns in the parietal lobes. These patterns reinforced earlier findings regarding the expansion of association cortices. These are areas responsible for integrating information from different senses.

Interhemispheric Connectivity

A study published in the journal *Brain* in 2013 provided further insight into how the two halves of Einstein's brain communicated. Weiwei Men, Dean Falk, and colleagues analyzed the corpus callosum. This structure is a thick bundle of nerve fibers that acts as a bridge connecting the left and right cerebral hemispheres.

The researchers used high-resolution photographs to measure the thickness of the corpus callosum at various points. They compared these measurements to MRI data from two control groups: fifteen elderly men and fifty-two young men.

The analysis revealed that Einstein's corpus callosum was thicker than that of the controls in most subregions. A thicker corpus callosum implies a higher number of nerve fibers. This suggests that Einstein possessed enhanced connectivity between the two hemispheres of his brain.

Increased connectivity allows for faster and more integrated processing of information. The authors proposed that this facilitated cooperation between the analytical functions often associated with the left hemisphere and the spatial or creative functions of the right hemisphere. This neural integration may have supported his ability to visualize complex physics problems.